1. AIM

To determine the I-V (voltage-current) characteristics of MOSFET (I_D vs. V_{DS}) using Keithley 4200A-SCS Parameter Analyzer.

2. APPARATUS / TOOLS REQUIRED

The following equipment and tools are required for the measurement:

• Keithley 4200A-SCS Parameter Analyzer: Core instrument with at least three Source-Measure Units (SMUs) for multi-terminal characterization.

• Probing Station / Test Fixture: Used for holding and contacting the device under test (DUT).

• Clarius+ Software: Keithley's measurement and analysis interface running on the connected system.

• NMOS Transistor (DUT): Device under test mounted on the test fixture.

• Low-Noise Triaxial / Coaxial Cables: Shielded interconnect cables to minimize noise during low-current measurements.

3. THEORY

Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET):

A MOSFET is a four-terminal semiconductor device consisting of Gate, Drain, Source, and Body (Bulk) terminals. The drain current (I_D) is modulated by the gate-to-source voltage (V_{GS}) across the gate dielectric. In standard NMOS characterization, the body terminal is tied to the source and held at ground potential.

Output Characteristics (I_D vs. V_{DS}):

The output characteristics describe the relationship between drain current (I_D) and drain-to-source voltage (V_{DS}) for different fixed gate-to-source bias voltages (V_{GS}).

When $V_{GS} < V_{th}$ (threshold voltage), the device is in the cut-off region and conduction current is negligible. When V_{GS} exceeds V_{th}, an inversion layer channel forms between source and drain.

The MOSFET operates across three distinct regimes:

1. Cut-off Region: $V_{GS} < V_{th}$; the channel is unformed, and drain current $I_D \sim 0$.

2. Linear (Triode) Region: At small V_{DS} ($V_{DS} < V_{GS} - V_{th}$), the inversion channel behaves as a voltage-controlled resistor where I_D increases linearly with V_{DS}.

3. Saturation Region: When $V_{DS} >= V_{GS} - V_{th}$, channel pinch-off occurs at the drain end, causing I_D to saturate and remain essentially constant with further increases in V_{DS}.

4. PROCEDURE

1. Power on the Keithley 4200A-SCS Parameter Analyzer and the connected computer interface.

2. Connect the analyzer via USB, LAN, or GPIB interface, and launch the Clarius+ software.

3. Verify that the hardware configuration detects all SMU modules properly.

4. Connect the SMU probes or triaxial cables to the MOSFET test fixture with correct terminal polarity (Gate, Drain, Source, Body).

5. In the Clarius+ library, select the standard FET I-V (vds-id) project template.

6. Assign SMU1 to Drain, SMU2 to Gate, and SMU3 / Ground to Source and Body.

7. Configure the drain voltage sweep: Set V_{DS} start, stop, and step values with appropriate current compliance limits.

8. Configure the gate voltage step list: Set stepping values of V_{GS} (e.g., 1 V to 4 V in 0.5 V or 1 V increments).

9. Execute the measurement scan and monitor the real-time I_D vs. V_{DS} curve acquisition.

10. Export the measured data points and save the generated characteristic family of curves.

11. Safely stop the test, discharge the device, disconnect test leads, and power down the equipment.

Note: Current compliance limits must be set appropriately before initiating sweeps to prevent thermal destruction of the DUT.

5. EXPERIMENTAL SETUP & MEASUREMENT RESULTS

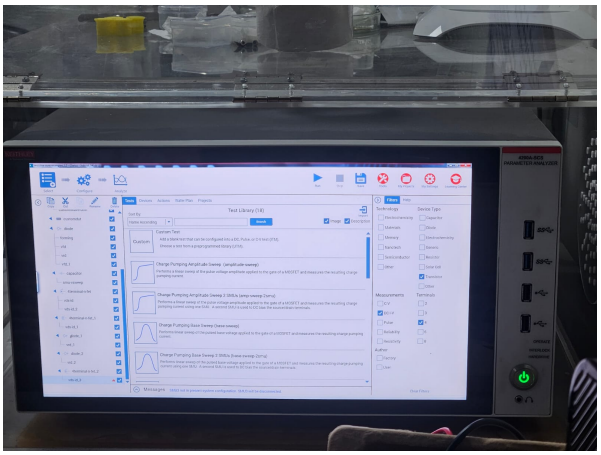


Figure 1: Keithley 4200A-SCS Parameter Analyzer

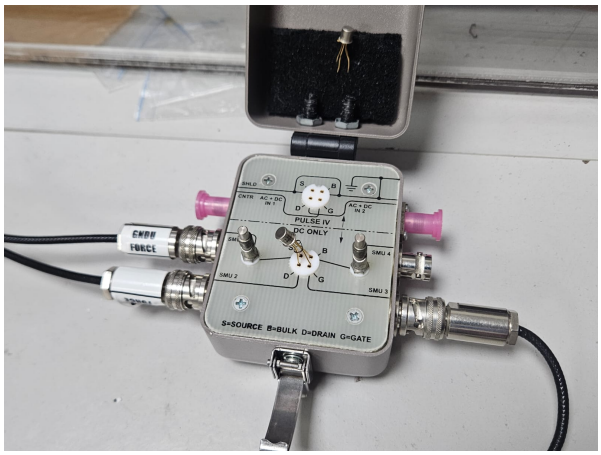


Figure 2: MOSFET Device Under Test (DUT) Fixture

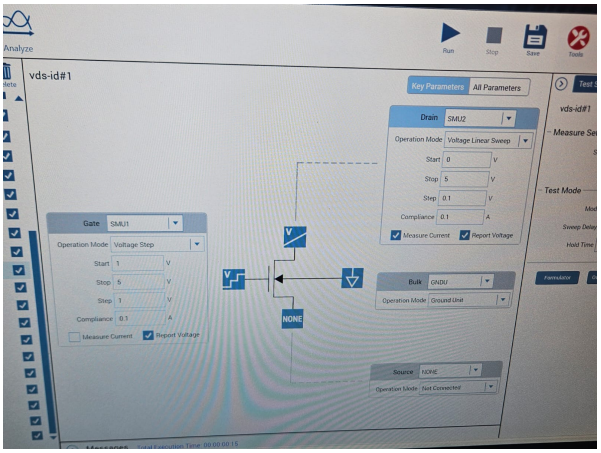


Figure 3: Clarius+ Voltage Sweep Configuration Interface

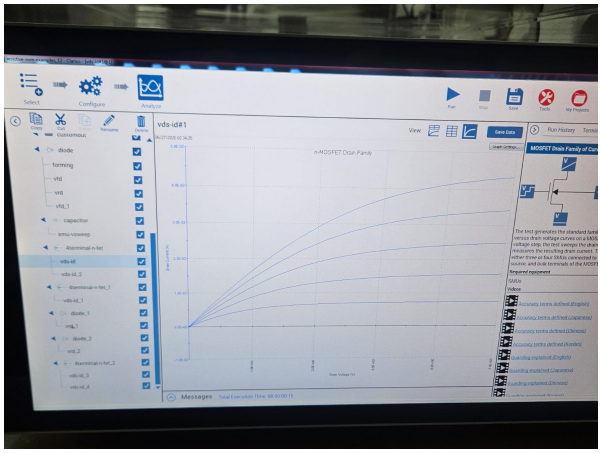


Figure 4: Measured MOSFET Output Characteristics (I_D vs. V_{DS})

6. OBSERVATION & CONCLUSION

Observations:

1. At small drain-to-source voltages (V_{DS}), the drain current (I_D) increases linearly, corresponding to the ohmic/linear regime.

2. Beyond the pinch-off point ($V_{DS} = V_{GS} - V_{th}$), the drain current levels off into saturation, demonstrating high output impedance.

3. Increasing the gate bias (V_{GS}) increases the channel carrier density, progressively shifting the saturation current upward.

4. A clean family of output characteristic curves was successfully recorded across all specified V_{GS} steps.

Conclusion:

The voltage-current (I_D vs. V_{DS}) output characteristics of an NMOS transistor were successfully measured and analyzed using the Keithley 4200A-SCS Parameter Analyzer. The linear, saturation, and cut-off operational regions were clearly demonstrated, confirming theoretical field-effect transistor operation.